

Learning Force-Aware Robot Manipulation with EMG-Labelled Human Data

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Abstract—We present LFEMG, a data-collection pipeline for using human demonstrations to support future training of force-aware robot policies. The pipeline collects RGB-D trajectories of humans manipulating objects in a calibrated camera frame while they wear a Myo armband that records electromyography (EMG) signals. Since contact force is not directly observable from RGB-D data alone, a central component of the project is learning to map EMG signals to grasping force, allowing human demonstrations to be augmented with force information.

To train and evaluate this mapping, we built a custom force-sensing object and designed complementary data-collection subtasks covering sustained grasping, no-contact behavior, and natural pick-and-place manipulation. Our EMG-to-force model achieves a test R^2 score of 0.84 across 20 held-out demonstration trajectories, suggesting that EMG can provide a strong proxy for contact force.

Finally, we integrated this force-labeling component with the broader robot data-collection pipeline by calibrating a camera with a set of xArm robots, implementing live-streaming code, and collecting real demonstrations. Together, these components show how human RGB-D and EMG data can be transformed into force-labeled demonstrations for future robot policy learning.

I. INTRODUCTION

A fundamental limitation of learning from human demonstrations is that force and contact dynamics are not directly observable from RGB input alone. While visual cues may reveal object motion and deformation, they do not explicitly encode grip force, internal muscle activation, or subtle contact transitions. This missing information creates an ambiguity gap between what the robot sees and the physical forces required to perform manipulation tasks.

Humans, in contrast, rely heavily on proprioception and muscle activation signals to regulate force during dexterous interaction. Electromyography (EMG) provides a direct measurement of muscle activity in the forearm and hand, offering a potentially rich modality correlated with grip force, contact onset, and manipulation intent.

This project investigates whether EMG signals can predict force-related information and estimate grasping force in human manipulation demonstrations. Our results show that EMG can recover meaningful force patterns across constant-hold, pick-and-place, and idle demonstrations, suggesting that EMG data can support future force-aware robot learning.

II. RELATED WORK

A. Force-aware learning and tactile sensing

In contact-rich manipulation, robust performance depends on reasoning about force, compliance, and slip. Learning-based systems demonstrate that policies augmented with tactile

sensing can achieve behaviors that are difficult or impossible for vision-only models, such as detecting incipient slip before object motion becomes visible, adapting grip force for fragile or deformable objects, and stabilizing in-hand manipulation through micro-adjustments based on contact feedback [8, 6].

While tactile sensing provides direct contact information, it requires instrumented robot hardware and does not address how force-relevant information can be incorporated when learning from scalable human demonstrations, which typically lack direct force measurements.

B. EMG as a proxy for force

Surface EMG measures muscle activation that directly drives grip force. Extensive studies show strong correlations between EMG amplitude and force magnitude, along with reliable decoding of grasp patterns and motor intent [2, 5, 4]. Large multimodal datasets demonstrate consistent relationships between EMG, hand kinematics, and manipulation behavior [1].

Recent multimodal models combine EMG and vision to improve grasp-type and intention recognition, and modern representation-learning approaches learn compact EMG embeddings suitable for downstream tasks [3, 9, 7]. However, prior work largely focuses on prosthetic control, gesture classification, or intent recognition. The use of EMG to provide force supervision for human demonstration data in robot learning remains relatively unexplored.

C. Positioning of this work

This project investigates whether synchronized EMG signals can serve as structured supervision to estimate force during human demonstrations. Similar to prior EMG-based work, we use muscle activity as a proxy for grasping effort. However, instead of focusing on gesture classification, our goal is to predict force trajectories that can label human demonstrations for future robot policy learning. Unlike tactile approaches that require instrumented robot hardware, EMG can be collected directly from human demonstrators, which makes data collection process simpler and more scalable.

III. METHODS

This project consists of two main components: training an EMG-to-force model and establishing a mapping from human hand motion to robot arm movement. Together, these components provide the inputs needed for downstream robot policy learning. Due to time constraints, we did not proceed to training the final robot policy. However, we completed the

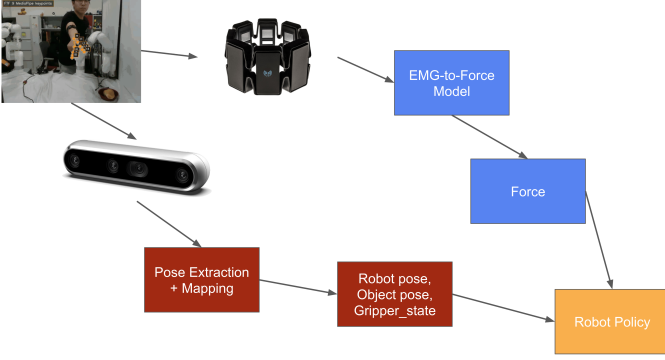


Fig. 1: Full LFEMG pipeline. During data collection, we capture synchronized EMG and RGB-D data. EMG is passed into a neural network to estimate contact force, while RGB-D data is processed into pose information and mapped to the robot frame. Both sources of information are used for downstream robot policy learning.

full pipeline, collected real demonstrations, and demonstrated how these data would support future policy learning. Below, Subsections A–E describe the EMG-to-force model, while Section F describes the human-hand-to-robot-arm mapping.

Figure 1 summarizes the full LFEMG pipeline. In this project, we complete the data-collection, EMG-to-force prediction, and hand-to-robot trajectory mapping components, which together produce force and motion labels for future downstream policy learning.

A. Hardware Components

1) **EMG Myo Armband:** The Myo Gesture Control Armband is made by Thalmic Labs, an 8 channel sensor to measure muscle activity on the forearm. We selected the Myo because we already have access to the device and interface stack that can be adapted to our purpose.

2) **SingleTact Capacitive Sensor:** The SingleTact sensor is an ultra-thin, single-element capacitive tactile sensor that measures applied force in a single axis. The sensor is read by a Seeed XIAO ESP32C6 running Arduino firmware, which polls the sensors over I2C and packages the readings into BLE notifications. We use SingleTact as our ground-truth force measurement for learning the EMG-to-force mapping.

3) **Final Hardware:** Figure 2 shows the custom hardware platform used for data collection. At a high level, a SingleTact force sensor was attached to a rigid block. The sensor was connected to an ADC, which converted the measured voltage signal into a digital value. The ADC was then wired to a Seeed Studio wireless board, which transmitted the data to our computer over Bluetooth.

To make the hardware more compact, we replaced the original breadboard setup by soldering the wiring onto a perfboard, allowing all components to fit directly on the block. We hypothesize that this block provides a reasonable approximation of other rigid objects with similar shape, mass,

and surface friction, making it sufficient for prototyping the data-collection pipeline.

B. Data Collection

The primary goal for data collection at this stage is to train the EMG-to-force neural network. During data collection, we collect 8 channels EMG signal read from the Myo armband, and the pairing ground-truth label is the force readings from the SingleTact. EMG streams at 200 Hz and force streams at 100 Hz.

1) **Datastream Synchronization:** Because this has evolved into a large stack, we introduced ROS2 to sync between hardware. When any of the signal (EMG, Force) is read into the PC, the corresponding driver stamps the sample with the current ROS2 clock and publish the sample to a unified data collector node, which reads the timestamp of all incoming data and compress them into zarr array.

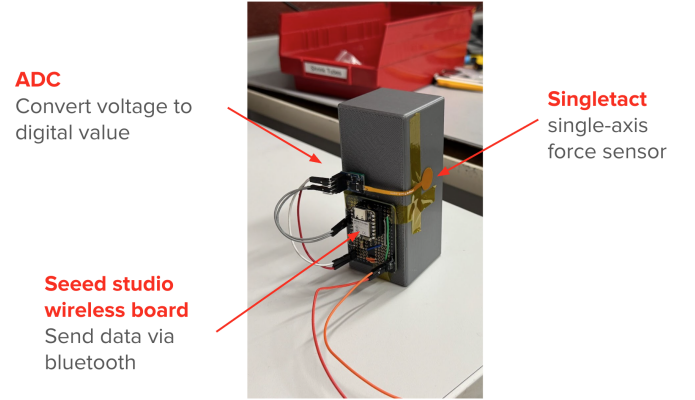


Fig. 2: Custom hardware we created to fit all modules compactly on a rigid block. This is a improved version from the bulky setup we had where all wires are connected to a breadboard.

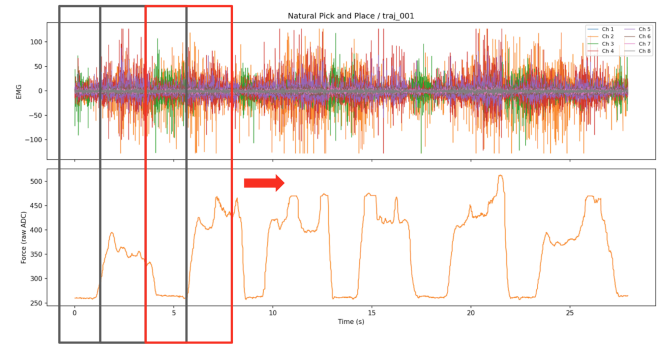


Fig. 3: EMG and SingleTact data synchronized and visualized. Top: EMG from Myo, all 8 channels depicted. Bottom: GT force label from SingleTact. We employ a sliding window of 200 EMG samples to prepare the data for learning.

We collected 300 demonstrations of pinching the rigid block between the thumb and index finger. Each trajectory spans

approximately 20 seconds and belongs to one of the following three tasks. The composition of our dataset is shown below:

- **33% Constant hold** - In a constant hold task, the subject grips the block and maintains an approximately fixed force throughout each trial, with the arm relaxed on the tabletop. The force should remain consistent within a trial, while the overall force level varies across trials. This provides a controlled and simplified learning setting, since the force output is temporally stable within each episode. We hypothesize that these steady-force trials help the model learn a reliable mapping from EMG activity to force magnitude alongside more dynamic scenarios.
- **34% Natural pick and place** - In a natural pick-and-place task, a typical trajectory involves approaching the object, picking it up, slowly ramping velocity from zero to a maximum and back down, and placing it back on the desk. Each sample may contain multiple such trajectories. Since the eventual action primitive of the robot involves pick and place, we want to "overfit" the model to recognize the associated scenarios it might encounter during real data collection for robot learning.
- **33% Idle** - In an idle task, the hand and arm remain relaxed, reaching, or hovering while exerting no force. These serve as negative examples to prevent the model from hallucinating force during non-grasping motion.

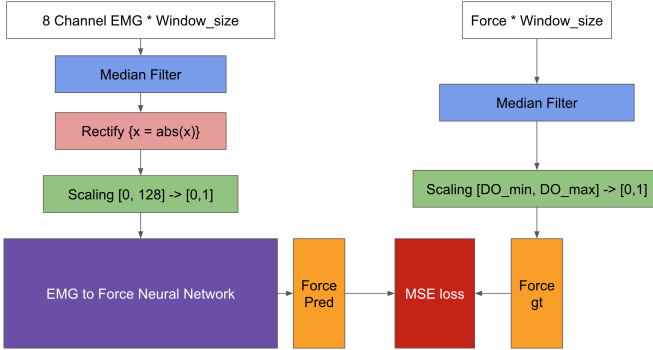


Fig. 4: Visualization of the full preprocessing pipeline.

C. Dataset Pre-Processing

Since each trajectory is roughly 20 seconds, at 200 Hertz streaming, that rounds up to 4000 EMG samples. Since force is streamed at 100 Hertz, we upsampled the force labels since we do not expect it to change in 10ms. Then, we use a sliding window of size 200 EMG samples with stride 20 across the trajectories to obtain paired samples for the Neural Network. Each training sample therefore consists of an EMG input window of shape 8×200 , paired with a corresponding force target sequence of length 200.

Raw EMG and force readouts are noisy and hard to work with. We conducted ablations and hyperparameter tuning and found the following procedure to work well. The procedure is also concisely depicted in Figure 3.

1) **Median Filter**: EMG signals were filtered using a 3-point median filter, while force signals were filtered using a 15-point median filter. The EMG filter was kept small to reduce noise while preserving fine temporal details. In contrast, heavier smoothing was applied to the force signal so that the model would learn the overall force trend rather than transient artifacts, such as brief drops caused by the demonstrator accidentally loosening their grip during data collection. This is reasonable because the robot is expected to maintain a relatively constant grasping force during manipulation.

2) **Rectify**: We take the absolute value for EMG since only magnitude of the signal matters for our purpose.

3) **Scaling**: Following prior work, we scaled both the input and output signals to the range $[0, 1]$ to improve training stability. In our experiments, this step was particularly important for the force signal, as the network often failed to converge reliably without output normalization.

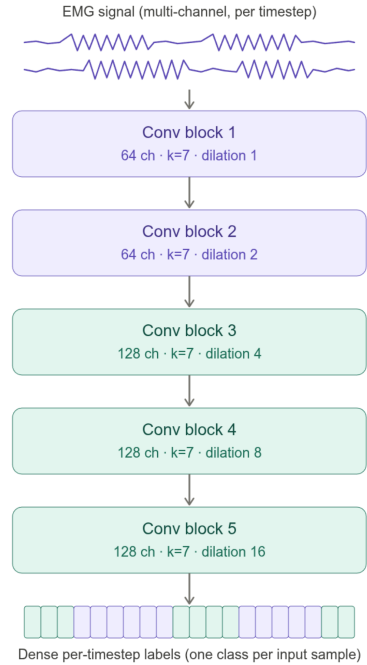


Fig. 5: EMG-to-force model parameterized by a neural network. We use a 1D CNN architecture with 5 Conv blocks. Each conv block consists of TemporalConv1D, GELU, and Dropout layers with residual connection.

To make better use of each recorded demonstration, we used dense supervision. Instead of assigning one label to an entire trajectory, we aligned the force signal with the EMG signal over time so that each 200-sample EMG window was paired with the corresponding 200-sample force trajectory. This produced timestep-level supervision from each demonstration and allowed the model to learn how force changes continuously with muscle activation.

D. Model Architecture and Training

We modeled EMG-to-force prediction as a temporal regression problem. The model takes an 8-channel EMG window as input and predicts a force trajectory over the same 200-timestep window. To capture temporal structure in the EMG signal, we used a Temporal Convolutional Network (TCN) composed of five one-dimensional convolutional blocks. The convolutional blocks used output channel sizes of 64, 64, 128, 128, and 128, respectively, with a kernel size of 7. We used increasing dilation factors of 1, 2, 4, 8, and 16, allowing the network to aggregate information over a wider temporal context without requiring a very deep architecture.

The model was trained in a non-causal setting, meaning that each prediction could use both past and future context within the input window. This was appropriate for our offline data-processing setting, where the goal was to learn accurate force labels from recorded demonstrations rather than deploy the model in real time. Each convolutional block used a GELU activation function and dropout with probability 0.15. Batch normalization was disabled in our final configuration.

The network was trained using mean squared error loss between the predicted and ground-truth force trajectories. We used the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . Training was performed with a batch size of 128 for up to 500 epochs. A cosine learning-rate scheduler was used over the full training horizon, and gradients were clipped to a maximum norm of 1.0 to improve training stability. We also used early stopping with a patience of 50 epochs based on validation loss. All experiments used a fixed random seed of 42 for reproducibility, and training metrics were logged using Weights & Biases.

E. Output Postprocessing

After model inference, we first unscaled the $[0, 1]$ mapping, and applied an additional smoothing step to the predicted force trajectory. In our final configuration, we used a Savitzky-Golay filter with a window length of 101 and polynomial order 2. This post-processing step reduced high-frequency fluctuations in the predicted force while preserving the overall shape of the force trajectory. We chose this over heavier median filtering because the downstream robot policy primarily needs a smooth and physically plausible force profile, but abrupt changes in the force trend should still be retained when they correspond to real changes in contact.

F. Camera-to-Robot Mapping and Pose Extraction

After developing the EMG-to-force model, we also need to map human hand motion into the robot frame so that the demonstrations can be used for downstream robot policy learning. To do this, we first performed hand-to-eye calibration between a RealSense D435 camera and the xArm setup. This calibration provides the transform ${}^B T_C$, which maps 3D points from the camera frame C into the robot base frame B .

We then extract human hand keypoints using MediaPipe Hands [10]. From the full set of detected hand landmarks, we select a nine-point subset covering the wrist and the two

fingers involved in the pinch grasp. These keypoints provide the geometric information needed to estimate the hand pose in the camera frame, denoted as ${}^C T_H$, where H represents the hand frame.

Finally, we convert the tracked thumb-index pinch geometry into a robot-frame end-effector target using the transform chain

$$p_B = {}^B T_C {}^C T_H p_H,$$

where p_H is the pinch target expressed in the hand frame, ${}^C T_H$ maps the hand frame to the camera frame, and ${}^B T_C$ maps the camera frame to the robot base frame. This produces a corresponding target position p_B in the robot base frame, which can then be used as an end-effector target for the xArm.

For each frame, the TCP position is defined as the midpoint between the thumb and index contact points, and the TCP orientation is estimated from the hand approach direction and pinch axis. The resulting pose is orthonormalized, converted to axis-angle form, temporally smoothed, and saved as a 6-D target pose $q_t = [x, y, z, r_x, r_y, r_z]$.

For gripper state, we use a binary command from the thumb-index pinch gap. For each frame, the distance between the thumb tip and index tip is thresholded to determine whether the gripper should be open or closed. The final output is a synchronized binary gripper command with the robot-frame TCP pose and EMG samples.

IV. EVALUATION

A. EMG-to-force

We evaluated the EMG-to-force model by comparing predicted force trajectories against ground-truth force measurements from the SingleTact sensor on held-out demonstration data. Because the predicted force signal is intended to provide supervision for downstream robot policy learning, we focused on whether the model could recover the overall temporal structure of the force trajectory, rather than only minimizing pointwise error.

Figure 7 shows representative prediction results across constant-hold, idle, and natural pick-and-place demonstrations. As a baseline, we trained a smaller version of the same temporal-convolutional architecture directly on raw EMG inputs and unprocessed force labels. The baseline predictions are noisy and often fail to follow the ground-truth force trend, especially in the idle and pick-and-place examples, suggesting that directly training on raw signals is insufficient for stable force prediction.

In contrast, the final model with preprocessing and post-processing produces substantially smoother and more physically plausible force trajectories. In the constant-hold example, the model captures the sustained grasping force as well as the temporary force drop near the middle of the trajectory. In the idle example, the final prediction remains close to the low-force ground-truth and avoids the large spurious spikes seen in the baseline. In the pick-and-place example, the final model follows the repeated force buildup and release patterns much more closely than the baseline, indicating that the model

learned meaningful temporal correspondence between EMG activity and grasping force.

The force values are reported in units of ADC digital output rather than Newtons. Since the digital output can be mapped to physical force through calibration, training on ADC output is equivalent to learning force up to a deterministic rescaling of the target values.

Quantitatively, our final model achieved an overall test R^2 score of 0.84 across 20 held-out demonstration trajectories. We used R^2 as the primary summary metric because it is dimensionless and measures the fraction of variance in the ground-truth force signal explained by the model, making it more interpretable than raw MAE or RMSE, which depend on the scale of the ADC digital output.

Among the three task types, the model performed best on natural pick-and-place trajectories. This is consistent with our design choice: pick-and-place is the primary downstream robot learning task, while the constant-hold and idle demonstrations were included mainly to provide complementary examples of sustained grasping and no-contact behavior. The idle task was the most challenging: although the target force should remain close to zero, the model must still be robust to EMG noise, sensor noise, small involuntary muscle activations, and other background variations.

We also computed MAE, RMSE, normalized RMSE, Pearson correlation, and cross-correlation-based metrics. However, these were less useful as aggregate measures because different task types have substantially different force distributions. Idle trajectories have low variance and nearly constant force, while natural pick-and-place trajectories contain repeated force buildup and release. For this reason, we report the overall test R^2 score together with representative predicted-versus-ground-truth trajectory plots.

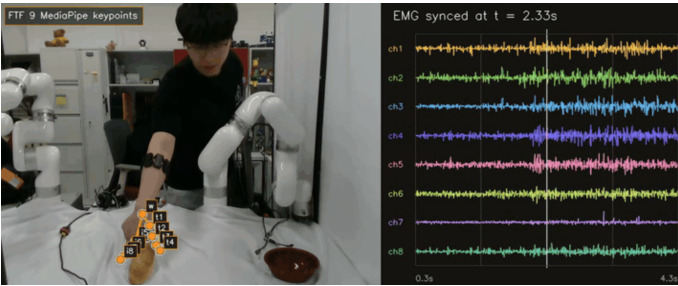


Fig. 6: Example synchronized trajectory for robot policy learning. The left panel shows human hand keypoints extracted with MediaPipe, while the right panel shows the corresponding EMG signals at the same timestep.

B. Full Robot Policy Pipeline

To demonstrate the feasibility of the full pipeline, we collected several real-world demonstrations for future policy learning. We implemented ROS2 nodes to synchronize data from the Myo armband and the RealSense camera, and saved the synchronized multimodal data into HDF5 files. Figure 6 shows an example of the running data-collection pipeline.

More specifically, each demonstration contains synchronized EMG and RGB-D streams. The EMG signal is passed through the trained EMG-to-force network to produce time-stamped force estimates. The RGB-D data is processed using the camera calibration and pose-extraction pipeline described in Section III-F, producing keypoint poses, object pose, and gripper-state information. In the final policy-learning setup, these processed demonstrations would be used to train a diffusion policy with a shifted time-horizon prediction objective. The policy would predict future robot end-effector poses, gripper states, and desired grasping forces.

At deployment time, the robot would use its current state, an external RGB-D camera for object pose estimation, and a force sensor mounted on the end-effector. The high-level diffusion policy would output the desired robot motion and grasping force, while a low-level force-feedback controller would adjust the gripper position to track the predicted force. In this way, the learned policy could use both visual information and force-aware supervision derived from human EMG demonstrations.

V. CONCLUSION

In this paper, we present LFEMG, a data-collection pipeline that combines RGB-D trajectories of human object manipulation with EMG signals to capture both visual and force-relevant information. Our final EMG-to-force model achieved an overall test R^2 score of 0.84 across 20 held-out demonstration trajectories, suggesting that EMG can recover meaningful grasping-force patterns from human demonstrations.

Compared with vision-only demonstration learning, LFEMG provides access to force information that is not directly observable from RGB-D data alone. Compared with tactile-sensing approaches, our method does not require the robot hand to be instrumented with dense tactile sensors during demonstration collection. Instead, EMG offers a lightweight way to capture force-relevant signals directly from the human demonstrator.

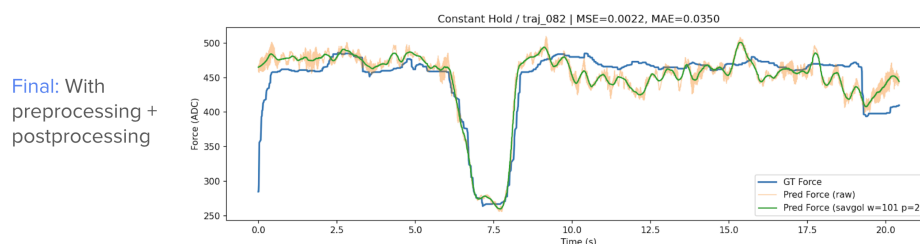
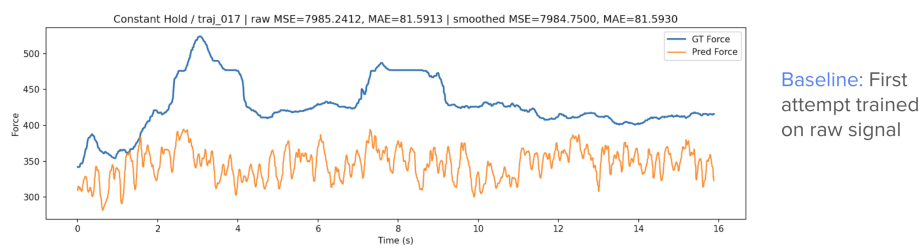
Future work should use the generated force and motion labels to train a downstream robot policy and evaluate whether EMG-labeled demonstrations improve real-world manipulation performance. In addition, ablation studies focusing on the EMG modality would help quantify its contribution. For example, future experiments could compare policies trained with predicted EMG-based force labels against policies trained from vision-only demonstrations, ground-truth force labels, or alternative force-estimation baselines.

VI. ACKNOWLEDGEMENT

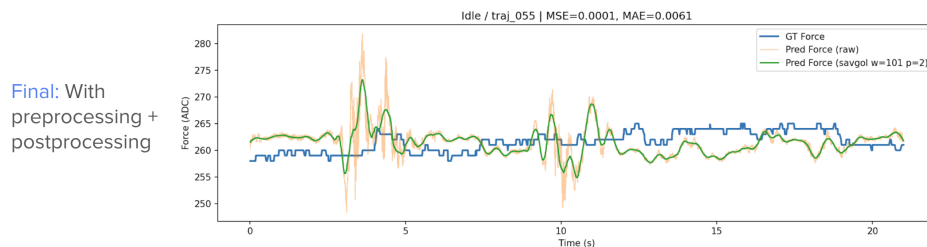
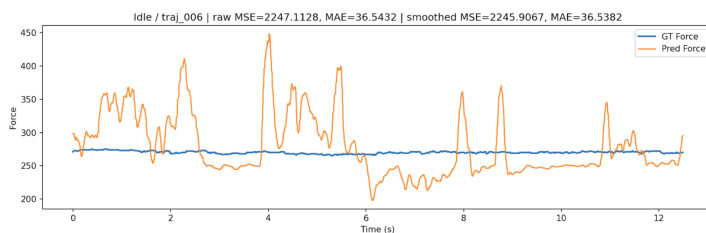
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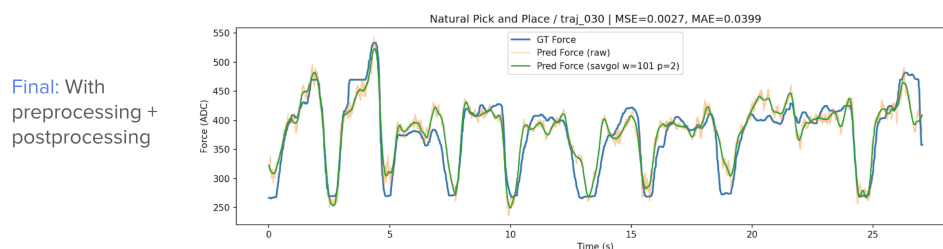
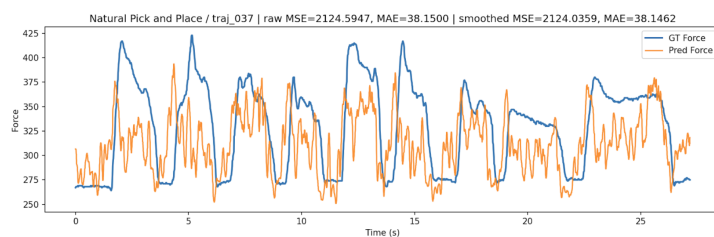
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(a) Constant hold



(b) Idle



(c) Pick-and-place

Fig. 7: Predicted and ground-truth force trajectories on representative held-out test demonstrations. Baseline results use raw EMG and force signals without the full preprocessing and post-processing pipeline, while final results use the complete pipeline.

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